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CATHETER AND HANDLE ASSEMBLY, SYSTEMS, AND METHODS

Abstract

A catheter and handle assembly for use in treating tissue. More specifically, the invention relates to a catheter assembly having an elongate shaft, ablation assembly coupled to one end of the shaft, and a handle coupled to the other end of the shaft. The handle facilitates both axial and circumferential positioning of the ablation assembly via the shaft in an airway, conduit, or vessel for treatment of the tissue. Embodiments include the use of a handle to facilitate the axial and circumferential positioning of the shaft and ablation assembly through and independent of a working channel of a bronchoscope during treatment for pulmonary disease like COPD and asthma.

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Background/Summary

RELATED APPLICATION [0001] The present application is a continuation of U.S. application Ser. No. 15/103,500, filed Jun. 10, 2016, which is a National Phase entry of PCT Application No. PCT/US2014/069971 filed Dec. 12, 2014, which claims the benefit of U.S. Provisional Application No. 61/915,282 filed Dec. 12, 2013, which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present invention relates generally to systems, apparatuses, and methods for treating tissue, and more particularly, the invention relates to a catheter and handle system including a catheter positioning handle assembly that facilitates both circumferential and axial positioning of a distal end of the catheter positioned in an airway, conduit, or vessel.

BACKGROUND

[0003] Pulmonary diseases are some of the most common medical conditions, affecting tens of millions of people in the U.S. alone. Pulmonary diseases result from problems in the respiratory tract that interfere with proper respiration. Many of these diseases require medical attention or intervention in order to restore proper lung function and improve a patient's overall quality of life. Some of the more common pulmonary diseases include asthma and chronic obstructive pulmonary disease or COPD. Symptoms of pulmonary disease like COPD and asthma vary but often include a persistent cough, shortness of breath, wheezing, chest tightness, and breathlessness. Generally, these symptoms are exacerbated when performing somewhat strenuous activities, such as running, jogging, brisk walking, etc.

[0004] However, these symptoms may be noticed when performing non-strenuous activities, if the disease is allowed to progress unchecked. Over time, especially if medical attention is not sought, a person's daily activities will be significantly impaired, thus reducing overall quality of life.

[0005] Many pulmonary diseases, whether acute or chronic, often involve pathologic conditions associated with airway inflammation. When such inflammation has developed at the airway, infiltrated inflammatory cells cause damage to bronchial or lung tissue, which eventually results in the respiratory dysfunction characteristic of pulmonary diseases, such as reduction in respiratory flow rate or oxygen exchange capacity. Over time, this inflammation can lead to blockage of the airway lumen, thickening of the airway wall, and alteration of structures within or around the airway wall. Airway obstruction can significantly decrease the amount of gas exchanged in the lungs resulting in breathlessness. Blockage of an airway lumen can be caused by excessive intraluminal mucus, edema fluid, or both. Thickening of the airway wall may be attributable to excessive contraction of the airway smooth muscle, airway smooth muscle hypertrophy, mucous glands hypertrophy, inflammation, edema, or combinations of these. Alteration of structures around

the airway, such as destruction of the lung tissue itself, can lead to a loss of circumferential traction on the airway wall and subsequent narrowing of the airway. Generally, pulmonary diseases like COPD and asthma are the result of a complex interplay of local inflammatory cytokines, inhaled irritants (e.g., cold air, smoke, allergens, or other chemicals), systemic hormones (e.g., cortisol and epinephrine), local nervous system input (i.e., nerve cells contained completely within the airway wall that can produce local reflex stimulation of smooth muscle cells and mucous glands), and the central nervous system input (i.e., nervous system signals from the brain to smooth muscle cells and mucous glands carried through the vagus nerve).

[0006] Asthma can further include acute episodes or attacks of additional airway narrowing via contraction of hyper-responsive airway smooth muscle that significantly increases airflow resistance. Asthma symptoms include recurrent episodes of breathlessness (e.g., shortness of breath or dyspnea), wheezing, chest tightness, and coughing. Additionally, COPD, often referred to as emphysema, is characterized by the alteration of lung tissue surrounding or adjacent to the airways in the lungs. Emphysema can involve destruction of lung tissue (e.g., alveolar sacs) that leads to reduced gas exchange and reduced circumferential traction applied to the airway wall by the surrounding lung tissue. The destruction of alveoli tissue restricts the in-flow of oxygen rich air and the proper function of healthier tissue, resulting in significant breathlessness. Exposure to chemicals or other substances (e.g., tobacco smoke) may significantly accelerate the rate of tissue damage or destruction. Additionally, chronic bronchitis, another type of COPD, is characterized by contraction of the airway smooth muscle, smooth muscle hypertrophy, excessive mucus production, mucous gland hypertrophy, and inflammation of airway walls. Like asthma, these abnormalities are the result of a complex interplay of local inflammatory cytokines, inhaled irritants, systemic hormones, local nervous system, and the central nervous system. Unlike asthma where respiratory obstruction may be largely reversible, the airway obstruction in chronic bronchitis is primarily chronic and permanent.

[0007] Treatment for pulmonary diseases includes reducing exposure to harmful agents, administering medications (e.g., bronchodilators, steroids, phosphodiesterase inhibitors, theophylline, antibiotics, etc.), administering lung therapy (e.g., oxygen therapy, pulmonary rehabilitation), and surgical intervention, such as bronchial thermoplasty. Unfortunately, pharmacological treatment requires patient compliance, often causes harmful side effects, and does not necessarily treat the underlying cause of the disease. Similarly, surgical intervention can result in the destruction of smooth muscle tone and nerve function, such that the patient is unable to respond favorably to inhaled irritants, systemic hormones, and both local and central nervous system input.

[0008] An alternative method for treating pulmonary disease is referred to as targeted lung denervation. This method utilizes ablation, such as RF ablation, via an ablation assembly to selectively treat target regions inside of the airway wall (e.g., anatomical features in the stromas) while protecting the superficial tissues, such as the surface of the airway wall. For example, the mucous glands can be damaged to reduce mucus production a sufficient amount to prevent the accumulation of mucus that causes increased air flow resistance while preserving enough mucus production to maintain effective mucociliary transport, if needed or desired. Nerve branches/fibers passing through the airway wall or other anatomical features in the airway wall can also be destroyed.

[0009] Specially designed catheters allow for the introduction of an ablation assembly, generally comprising one or more collapsible electrodes or energy emitters, coupled to an expandable member, such as a balloon, into the airway of a patient via a delivery device. The delivery device can be a guide tube, a delivery sheath, a bronchoscope, or an endoscope and can include one or more viewing devices, such as optical viewing devices (e.g., cameras), optical trains (e.g., a set of lens), optical fibers, CCD chips, and the like. Once positioned in the desired region of the airway, such as the left and/or right main bronchi, the expandable member is expanded to position the one

or more electrodes in contact with the airway wall.

[0010] Energy, such as RF energy, is supplied to the energy emitter to ablate the targeted tissue, causing a lesion to form, therefore temporarily or permanently damaging the targeted tissue, therefore affecting, e.g. attenuating nerve signals to or from, portions of the lungs associated with the targeted tissue. Simultaneously, a coolant is supplied through the catheter and is directed to the one or more electrodes and into the expandable member or balloon. This allows for cooling of the superficial tissue in contact with the electrode, as well as the adjacent tissues. The size, shape, and depth of the lesions are determined by the flow rate and temperature of the coolant, and the energy supplied to the energy emitter(s). Devices, systems, and methods of such procedures can be found, for example, in one or more of U.S. Pat. No. 8,088,127 entitled “Systems, Assemblies, and Methods for Treating a Bronchial Tree,” and U.S. Patent Application Publication No. 2011/0152855 entitled “Delivery Devices with Coolable Energy Emitting Assemblies,” both of which are incorporated herein by reference in their entireties.

[0011] In order to ensure that most or all of the target nerves extending along the airway are treated, it is generally desirable to form a circumferential lesion around all or most of the circumference of the airway wall. Due to design constraints or preferences, the electrode or energy emitter may not extend around the entirety of the circumference of the airway wall. Therefore, a circumferential lesion may be formed by ablating tissue while slowly rotating the ablation assembly or by positioning the ablation assembly in a series of rotational positions at each of which energy is delivered for a desired time period. The adjacent lesions then become contiguous and form a circumferential band all the way around the airway wall. Additionally or alternatively, the catheter may be repositioned axially to treat other locations within the airway distally or proximally of the first treatment site.

[0012] Typically targeted lung denervation will be performed under bronchoscopic visualization. A bronchoscope may be introduced into the target airway and the treatment catheter then delivered either alongside the bronchoscope or, more preferably, through the working channel of the bronchoscope. However, placement through the working channel can create challenges in manipulating the catheter due to the small size of the working channel, friction between the catheter and the walls of the working channel, and, in the case of flexible bronchoscopes, the curvature or tortuosity of the working channel. Furthermore, it is easy to lose the relative position of the electrode from the camera if both can rotate independently because the camera will always project right side up in the airway, regardless of the position of the bronchoscope.

[0013] Disorientation of the electrode in the treatment site and/or unintended movement of the catheter relative to the bronchoscope during treatment may result in treatment inaccuracies, causing axial or circumferential misalignment of lesions, undesirable gaps between lesions, or excessive overlap between lesions.

[0014] To address these and other challenges, there remains a need for a system, device, or apparatus for precise positioning and manipulation of pulmonary treatment catheters, such as targeted lung denervation catheters, while easily maintaining electrode orientation when positioned in a pulmonary airway through a delivery device such as the working channel of bronchoscope.

SUMMARY

[0015] Embodiments of the invention are directed to a pulmonary treatment catheter and handle system including a catheter assembly, a handle assembly, and a scope coupling assembly for coupling the handle assembly and catheter assembly to a delivery device, such as a bronchoscope. Embodiments are further directed to kit including the catheter assembly and handle assembly for coupling together and to a delivery device, such as a bronchoscope, via the scope coupling assembly, and instructions for methods of using such. The catheter assembly is further fluidly and electrically coupled to a system console, including a coolant supply and return reservoir, and an energy supply such as a RF generator, via the handle assembly.

[0016] The catheter assembly, handle assembly, and scope coupling assembly cooperate together to

facilitate both circumferential and axial positioning of a catheter electrode in a treatment site, such as an airway, conduit, or vessel for treatment of the tissue, while maintaining known rotational and axial orientation of portions of the catheter assembly, such as an ablation assembly including an energy emitter or electrode, within the treatment site. The system can further facilitate optical coupling of the ablation assembly of the catheter assembly with a viewing device, such as a fiber optic camera at a working end of a bronchoscope, while maintaining independent movement of the viewing device with respect to the ablation assembly to achieve maximum viewing flexibility of the treatment site and ablation assembly. This allows for full viewing access of the electrode of the ablation assembly within the treatment site regardless of its orientation or positioning within the treatment site.

[0017] In embodiments, the catheter assembly comprises a targeted lung denervation RF, microwave, or ultrasound catheter, and generally includes an elongate shaft, and an ablation assembly coupled to a distal portion of the shaft, the ablation assembly including an expandable member, such as a balloon or basket, and one or more electrodes or energy emitters coupled to the expandable member. The catheter assembly also includes a cooling circuit including a coolant inflow and outflow lumen within the elongate shaft, and a coolant inlet path and return path (e.g. cooling conduit(s)) to circulate coolant to the expandable member and to the energy emitter, one or more power wires for supplying power to the energy emitter, optional thermocouple(s) and associated wires for measuring and sensing temperature at locations proximal to the electrode, optional cooling circuit pressure sensors and associated wires for measuring and sensing pressure within the cooling circuit, and/or optional pressure relief valves.

[0018] In embodiments, the handle assembly is coupled to a proximal portion of the shaft. The handle assembly can include a housing fixedly coupled to the proximal end of the shaft, and a spindle tube or handle frame coupled to the housing such that the spindle tube is rotatably and axially shiftable with respect to the housing and the catheter assembly. The handle assembly can further include an umbilical cable with strain relief for coupling the handle assembly, and ultimately the catheter assembly, to a system console including a heat exchanger, coolant pump, energy generator (such as an RF, microwave, or ultrasound generator), and a system controller. The umbilical cable can aid in coupling, for example, inlet and return fluid tubes (coolant) from the system console for fluidly coupling the catheter assembly to the heat exchanger and pump of the console, an electrical cable/connector to electrically connect the electrode of the catheter assembly to an energy source, thermocouple wires to monitor temperature of the surface tissue of the treatment site, the electrode, or both, and/or pressure sensors to monitor the high pressure coolant inlet flow and the low pressure return flow.

[0019] In embodiments, the scope coupling assembly removably couples the handle assembly and catheter assembly to a working channel of a delivery device, such as a bronchoscope, in a unique or single orientation, such that the initial rotational and axial orientation of the catheter assembly with respect to the working end or tip of the bronchoscope is known. The coupling assembly is fixedly coupled to the spindle tube, such that the handle housing and catheter assembly are rotatably and axially shiftable with respect to the working channel.

[0020] In some embodiments, the handle assembly is configured to maneuver the distal portion of the catheter shaft having the ablation assembly thereon with respect to the delivery device and airway wall, in axial and circumferential directions during or in preparation for the administration of treatment for a pulmonary disease. In some aspects, the delivery device is a bronchoscope that comprises a working channel or port through which the elongate shaft and ablation assembly of the catheter can be inserted, and the handle assembly can be secured to the bronchoscope in a manner that functionally engages the handle assembly movements with the bronchoscope and the ablation assembly of the catheter assembly. For example, the handle assembly can be fixed to the bronchoscope such that the shaft of the catheter assembly, and therefore the ablation assembly, rotates and axially translates with corresponding rotational and axial translation of the

bronchoscope for coarse adjustment of the ablation assembly within the airway. The handle assembly may also include one or more steering or manipulation mechanisms that are functionally coupled with the shaft and/or ablation assembly to translate the shaft and/or ablation assembly rotationally and/or axially with respect to the scope for either coarse or fine circumferential and/or axial adjustment while other portions of the handle assembly remain engaged with and stationary relative to the bronchoscope.

[0021] In a particular embodiment, a length of the elongate shaft of the catheter assembly and the manipulation mechanism of the handle assembly are configured to allow for a travel length of the ablation assembly to be equal to or greater than a longitudinal length of the ablation assembly such that an axial stroke of the handle assembly (fully retracted to fully extended) allows the ablation assembly to shift between being fully extended out of the working end or tip (with or without a gap outside of the working end) of the bronchoscope to fully retracted within the working end of the bronchoscope (with or without a gap inside of the working end).

[0022] In some embodiments, once engaged with the bronchoscope, a user can maneuver the bronchoscope, and the elongate shaft and ablation assembly thereon, into a position advantageous for conducting treatment for a pulmonary disease (e.g., a targeted lung denervation procedure). In some embodiments, the handle assembly can comprise controls for independently adjusting the distal portion of the elongate shaft of the catheter and/or the ablation assembly from the bronchoscope. In yet another embodiment, the handle assembly can comprise independent control mechanisms for axial translation of the catheter assembly and rotational movement of the catheter assembly with respect to the working channel of the bronchoscope.

[0023] In use, the handle is permanently or temporarily coupled to the catheter assembly, and the distal portion of the catheter shaft and ablation assembly are inserted into the working channel or port of a delivery device, such as a bronchoscope. The handle assembly is then coupled to the bronchoscope via the coupling assembly in a unique orientation. In some embodiments, the distal portion of the delivery device can comprise a visualization device, a camera, and/or an aspiration lumen or vacuum. In some embodiments, the distal end of the catheter shaft and ablation assembly can be positioned using the axial and circumferential controls of the handle assembly, such that a specific portion of airway tissue can be targeted for treatment (e.g., radio frequency energy emission from an electrode). In embodiments, the working end of bronchoscope can be maneuvered independently of the catheter shaft and ablation assembly. For example, the handle assembly can be used to more accurately position an electrode of the ablation assembly based on visual cues from the camera of the delivery device, without moving the delivery device and camera.

[0024] In embodiments, the distal end of the catheter assembly includes a longitudinal indicator band or stripe. When the catheter assembly and handle assembly are coupled to the working channel of the bronchoscope, the indicator band aligns with the center point of the camera of the bronchoscope. This band provides a visual confirmation of the ablation assembly orientation within the treatment site, and regardless of the rotational orientation of the bronchoscope within the treatment site, the location of the ablation assembly with respect to the working end of the bronchoscope is known. Furthermore, the band provides a visual confirmation of the axial location of the ablation assembly with respect to the working end of the bronchoscope to provide optimal optical coupling of the bronchoscope camera and the ablation assembly.

[0025] The system according to embodiments allows for one-handed operation of both the catheter assembly and the bronchoscope when rotated and/or translated axially together for coarse adjustment, as well as for the independent rotationally and axial fine adjustment of the ablation assembly with respect to the bronchoscope.

[0026] The above summary of the various representative embodiments of the invention is not intended to describe each illustrated embodiment or every implementation of the invention. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and

understand the principles and practices of the invention. The figures in the detailed description that follow more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The invention can be more completely understood in consideration of the following detailed description of various embodiments of the invention in connection with the accompanying drawings, in which:

[0028] FIG. 1 is an illustration of a pulmonary treatment catheter and handle system including a catheter assembly having an elongate shaft and an ablation assembly coupled to one end of the shaft, and a handle assembly coupled to the other end of the shaft, according to an embodiment.

[0029] FIG. 2 is a top view of a pulmonary treatment catheter and handle system including a catheter assembly having an elongate shaft and an ablation assembly coupled to one end of the shaft, and a handle assembly coupled to the other end of the shaft, according to an embodiment.

[0030] FIG. 3 is an illustration of an ablation assembly coupled to the distal portion of the elongate shaft of the catheter according to an embodiment.

[0031] FIG. 4 is a diagram illustrating the direction of fluid flow in a cooling circuit according an embodiment.

[0032] FIG. 5A-C is an illustration of three positions of a catheter and handle assembly coupled to a delivery device according to an embodiment.

[0033] FIG. 6 is an exploded view of a handle assembly and coupling assembly according to an embodiment.

[0034] FIG. 7 is a partial cutaway view of the handle assembly of FIG. 4 in a retracted or closed configuration.

[0035] FIG. 8 is a cross-sectional view of an elongate shaft of a catheter according to an embodiment.

[0036] FIG. 9 is a cross-sectional side view of a manifold coupled to a catheter shaft, catheter tube, and an umbilical cable according to an embodiment.

[0037] FIG. 10 is a perspective and partially exploded view of manifold and catheter shaft according to an embodiment of the invention.

[0038] FIG. 11 is a cross-sectional side view of a manifold coupled to a catheter shaft and an umbilical cable according to an embodiment.

[0039] FIG. 12 is a perspective view of a manifold according to an embodiment.

[0040] FIG. 13 is a cross-sectional side elevational view of a catheter and handle system with a pressure monitoring assembly according to an embodiment.

[0041] FIG. 14 is a cross-sectional side elevational view of a catheter and handle system with a pressure monitoring assembly according to an alternative embodiment.

[0042] FIG. 15 is a cross-sectional side elevational view of a catheter and handle system with a pressure monitoring assembly according to another alternative embodiment.

[0043] FIG. 16 is an end view of a working end of a bronchoscope with catheter extending through a working channel according to an embodiment.

[0044] FIG. 17A-C is an elevational, cross-sectional, and exploded illustration of a distal end of an elongate shaft of a catheter assembly having an orientation indicator thereon.

[0045] FIG. 18 is an illustration of the axial and circumferential controls of a handle assembly according to an alternative embodiment.

[0046] FIG. 19 is an illustration of the handle assembly wherein the axial and circumferential controls are decoupled according to an alternative embodiment.

[0047] FIG. 20A is a perspective view of a scope adapter assembly including a shroud and a

coupler according to an embodiment.

[0048] FIG. 20B is an exploded view of the adapter assembly of FIG. 20A.

[0049] FIG. 20C is a cutaway view of the adapter assembly of FIG. 20A, installed on a delivery device.

[0050] FIG. 20D is a perspective view of an alternative coupler, according to another embodiment.

[0051] FIG. 20E is a perspective view of an adapter assembly and delivery device in a first position, according to an embodiment.

[0052] FIG. 20F is a perspective view of the adapter assembly of FIG. 20E attached to the delivery device.

[0053] FIG. 21 is a schematic representation of a kit according to an embodiment.

DETAILED DESCRIPTION

[0054] As illustrated in FIGS. 1 and 2, a catheter and handle system **100** according to an embodiment of the present invention can comprise an ablation catheter assembly **101** having an elongate shaft **102** and an ablation assembly **106** coupled to a first or distal end of shaft **102**, a positioning handle assembly **104** coupled to a second or proximal end of shaft **102**, and a scope coupling assembly **103** for coupling catheter assembly **101** and handle assembly **104** to a working channel of a delivery device, such as a bronchoscope. Catheter assembly **101** is further fluidly and electrically coupled to a system console (not shown), including a coolant supply and return reservoir, and an energy supply such as a RF generator, via handle assembly **104**. Handle assembly **104** is configured to maneuver the distal portion or end of shaft **102** and therefore ablation assembly **106** in axial and circumferential directions during the administration of treatment.

[0055] As shown in FIG. 3, in one non-limiting embodiment of the invention, ablation assembly **106** can comprise one or more energy emitters **510**, such as an electrode, and an expandable member **520**, such as a balloon or basket. In some embodiments, ablation assembly **106** can comprise a coolant fluid path or cooling circuit **600** to cool electrode **510** and a surface of expandable member **520** to protect surface tissue in contact with electrode **510** and adjacent to electrode **510**. As depicted in FIG. 4, cooling circuit **600** includes coolant supplied from a reservoir **601** of a system console, through an optional heat exchanger **603** of the system console, through handle **104**, through an inflow lumen in shaft **102**, through conduit **540** to which electrode **510** is coupled, through expandable member **520**, through an outflow lumen in shaft **102**, through handle **104**, and back to the system console. Non-limiting examples of a system console can be found in U.S. Patent Application Publication No. 2013/0289556, entitled “Delivery Devices with Coolable Energy Emitting Assemblies” and U.S. Pat. No. 8,489,192 entitled “System and Method for Bronchial Dilation,” both of which are incorporated by reference in their entireties. Cycling of the fluid is accomplished, for example, by a peristaltic pump **605**. In an alternative embodiment, the flow is reversed such that the coolant flows through the expandable member before the electrode.

[0056] Referring back to FIG. 3, ablation assembly **106** can optionally comprise a throttle valve **530** for regulating the flow between conduit **540** and expandable member **520**. Ablation assembly **106** can also optionally comprise a support wire **1214**, such as a Nitinol wire, extends along at least a length of catheter shaft **102** and between an interior of a proximal end **520a** and distal end **520b** of expandable member **520** to provide added axial, torsional, and buckling support for expandable member **520** and catheter shaft **102**. More particularly, a first portion or end of **1214a** of support wire **1214** is coupled to the distal end **102a** of catheter shaft **102**, while a second end **1214b** is coupled to the distal end **520b** of expandable member **520** at a junction between throttle valve **530** and expandable member **520**.

[0057] Further details of the ablation assembly are described in U.S. Pat. Nos. 8,088,127 entitled “Systems, Assemblies, and Methods for Treating a Bronchial Tree,” and U.S. Patent Application Publication No. 2011/0152855 entitled “Delivery Devices with Coolable Energy Emitting Assemblies,” both of which were incorporated by reference in their entireties above.

[0058] In some embodiments, and referring back to FIGS. 1 and 2, handle assembly **104** can

comprise an umbilical cable **120** coupled to an end of handle assembly **104** via strain relief **121** for fluidly and/or electrically coupling catheter assembly **101** to accompanying devices or accessories, such as a power source, energy source, fluid or coolant supply, heat exchanger, and controller, preferably combined in a system console. Umbilical cable **120** can include, for example, connections for inlet and return fluid tubes or lumens **105** for fluidly coupling shaft **102** to a fluid or coolant supply, from the console which optionally includes a heat exchanger for cooling and/or heating input fluid, and one or more electrical cable/connector **107** to electrically connect the shaft and/or ablation assembly to a power source, thermocouples for temperature monitoring, and/or pressure sensors for coolant circuit pressures. In other embodiments, handle assembly **104** can comprise an internal battery source for operating handle assembly **104** and any accompanying devices or accessories.

[0059] Referring to FIG. 5, catheter assembly **101** and handle assembly **104** are capable of coupling to a delivery device **200**, such as, for example, a guide tube, a delivery sheath, a bronchoscope, or an endoscope. Delivery device **200** can include one or more viewing devices, such as optical viewing devices (e.g., cameras), optical trains (e.g., a set of lens), and the like. In one particular embodiment, delivery device **200** comprises a flexible bronchoscope. Ablation assembly (not shown) and elongate shaft **102** are inserted into a working channel port **202** of device **200**. Handle assembly **104** is then secured to device **200** via a scope coupling assembly **103**. Scope coupling assembly **103** can be integral with or coupled to handle assembly **104**, or can be its own stand-alone adapter coupling to both handle assembly **104** and port **202**. Scope coupling assembly **103** securely fits to port **202**, such as by frictional or abutting fit, locking lever, threaded engagement of corresponding threads, bayonet or snap fit, spring loaded fit, or any of a variety of mechanisms known to those skilled in the art. Scope coupling assembly **103** further can include a locking mechanism **208** such as a lever or slide lock to prevent unintended disengagement of handle assembly **104** from port **202**. Once secured, scope coupling assembly **103** is fixed axially and rotationally with respect to device **200**. If not previously assembled, handle assembly **104** is then secured to scope coupling assembly **103**, as will be described in more detail below.

[0060] In a particular embodiment of handle assembly **104** and referring to FIGS. 6 and 7, handle assembly **104** generally comprises a handle housing or cover **1002** thereby defining an interior space of handle assembly **104** having a longitudinal axis A, and a handle frame, such as a spindle tube **1024**. In an embodiment, housing **1002** comprises a first handle housing **1002a** and second handle housing **1002b** coupled to first handle housing **1002a** via one or more a mortise and tenon joints, corresponding male and female interlocking parts, screws, adhesive, welding, or any of a variety of mechanisms for joining the housings. In other embodiments not shown, the housing is monolithic, or alternatively, formed of more than two pieces.

[0061] A portion of elongate shaft **102** of the catheter assembly **101**, extends along longitudinal axis A within handle housing **1002**. A catheter tube **1004** sheaths and protects shaft **102** within housing **1002**. In an embodiment, and referring to FIG. 8, catheter shaft **102** includes a central coolant inlet lumen **102a** for supplying fresh or recycled coolant from a coolant supply of the system console to the ablation assembly coupled to a distal end of catheter **102** (e.g. shown in FIG. 6). A coolant return lumen **102b** surrounds coolant inlet lumen **102a** such that the coolant return lumen **102b** and coolant inlet lumen **102a** are coaxial.

[0062] However, other arrangements that are not coaxial, such as side-by-side, can also be contemplated.

[0063] Referring to FIG. 6, an umbilical cable assembly **1006** of handle assembly **104** at a proximal end couples coolant inlet lumen **102a** and coolant return lumen **102b** of shaft **102** to a coolant supply source and return reservoir. As explained previously with respect to FIGS. 1 and 2, umbilical cable assembly **1006** can couple any of a variety of tubes and/or electrical wires or cables to catheter tube **1004** and/or handle assembly **104** for fluidly and/or electrically coupling catheter shaft **102** and ultimately ablation assembly **106** to accompanying devices or accessories, such as a

power source, energy source, fluid or coolant supply, heat exchanger, and controller, preferably combined in a system console. Preferably, umbilical cable assembly **1006** comprises flexible umbilical cable **120** terminating in a rubber strain relief **121**.

[0064] Referring to FIGS. **9-12**, catheter tube **1004** and the proximal end of shaft **102** are operably coupled to umbilical cable assembly **1006** via a manifold **1008**. Manifold **1008** distributes or redirects the inflow, outflow, and/or electrical flows between shaft **102** and the system console (not shown).

[0065] In an embodiment, and referring to FIGS. **11** and **12**, manifold **1008** comprises a first inlet port **1008a** and a second inlet port **1008b**. First inlet port **1008a** and second inlet port **1008b** are in fluid communication with each other via conduits **1009a** and **1009b** and gate **1111**. A proximal end of coolant supply tube **1006a** of umbilical cable assembly **1006** extends within and is bonded to an interior surface of first port **1008a** of manifold **1008** by adhesive bonding, heat sealing, ultrasonic welding, or any of a variety of attachment mechanisms. A distal end (not shown) of coolant supply tube **1006a** is coupled to a coolant supply, optional heat exchange, and pump of the system console.

[0066] Inlet lumen **102a** and outlet lumen **102b** of shaft **102** extends within second port **1008b** of manifold **1008**, and outlet lumen **102b** is bonded to an interior surface of port **1008b** by adhesive bonding, heat sealing, ultrasonic welding, or any of a variety of attachment mechanisms. Inlet lumen **102a** extends beyond outlet lumen **102b** of shaft **102** through conduits **1009a** and **1009b**, and into coolant supply tube **1006a**. Optionally, a diameter of conduit **1009b** is similar to a diameter of inlet lumen **102a**, such that inlet lumen **102a** can be bonded to an interior surface of conduit **1009b** by adhesive bonding, heat sealing, ultrasonic welding, or any of a variety of attachment mechanisms.

[0067] Manifold **1008** further comprises a third port **1008c**. A proximal end of coolant return tube **1006b** of umbilical cable assembly **1006** extends within port **1008c** and is bonded to an interior surface of third port **1008c** by adhesive bonding, heat sealing, ultrasonic welding, or any of a variety of attachment mechanisms. Third port **1008c** is in fluid communication with second port **1008b**, and therefore outlet lumen **102b** secured thereto, via conduits **1009c** and **1009a**, separated by gate **1111**. A distal end (not shown) of coolant return tube **1006b** is coupled to a reservoir of the system console for recycle and/or disposal of coolant.

[0068] In use, coolant is supplied from the coolant supply of the system console through coolant supply tube **1006a** of umbilical cable assembly **1006**. Coolant then flows from coolant supply tube **1006a** into inflow lumen **102a** of catheter shaft **102**. Coolant flows along the length of shaft **102** into ablation assembly **106**, as described below. The coolant circulates through ablation assembly **106** and into outlet lumen **102b**, along the length of shaft **102** into manifold conduit **1009a**, reservoir, **1111**, and conduit **1009c**. Coolant flows out of manifold **1008** via coolant return tube **1006b** and back to the system console for recycle and/or disposal.

[0069] Referring back to FIGS. **9** and **10**, catheter tube **1004** is bonded to an exterior surface of second port **1008a** of manifold **1008**, thereby covering second port **1008a** and the portion of shaft **102** extending within and along handle assembly **100**. Catheter tube **1004** is secured to manifold **1008** by adhesive bonding, heat sealing, ultrasonic welding, or any of a variety of attachment mechanisms.

[0070] Referring to FIGS. **9**, **11**, and **12**, manifold **1008** can further comprise a recessed seat **1113** formed in a surface thereof for receiving and securing one or more printed circuit boards thereto, such as by soldering or additive printing directly thereon, and one or more liquid pressure sensors and gauges, flow sensors and gauges, and/or thermocouples for detecting and optionally displaying pressures, coolant inlet and outlet flow rates, and/or temperatures at various locations in the catheter assembly **101** and/or ablation assembly **106**.

[0071] Leads from the pressure sensor, flow sensors, and/or thermocouples are then fed through an aperture **1115** formed in seat **1113** of manifold **1008** which fluidly couples seat **1113** with gate **1111**. The leads extend through shaft **102** to ultimately reach ablation assembly **106**. In an embodiment,

and referring back to FIG. 8, a first lead **1010a** and a second lead **1010b** are fed through outlet lumen **102b** of shaft **102** to ablation assembly **106**.

[0072] Manifold **1008** is fixedly coupled to an interior of first and/or second handle cover **1002a,b** such that manifold **1008** and therefore catheter tube **1004** are fixed axially and rotationally to covering **1002**. Manifold **1008** can be fixed to handle cover **1002a** and/or **1002b** by insertion of a peg formed on the interior of housing **1002a** or **b** into a corresponding sleeve **1117** formed on manifold **1008** as seen in FIG. 12, for example. However, any of a variety of attachment mechanisms can be contemplated.

[0073] Thermocouples can be placed anywhere within the catheter assembly **101** including at the energy emitter or electrode site of the ablation assembly **106**, within the cooling member of the ablation assembly to measure the temperature of the coolant in the cooling member, and/or within the inflow and/or outflow lumens **102a**, **102b** in the elongate shaft **102** to measure the temperatures of the inflow and/or outflow coolant flows. Coolant temperature and/or coolant flow are then either manually or automatically regulated at the system console to ensure that the ablation site is sufficiently heated to cause permanent damage to target tissue, while ensuring that surface tissue in contact with the electrode and cooling member is sufficiently cooled to inhibit or prevent permanent damage to the surface tissue. Pressure and/or flow rate can be measured within the inflow and/or outflow lumens, and/or within the cooling member and/or ablation assembly to measure theoretical and/or direct pressure of the ablation assembly. Pressure within the inflow lumen can be high compared to pressure within the outflow lumen due to a throttle positioned within the ablation assembly. In a particular embodiment, a pressure of the expandable member, e.g. balloon, of the ablation assembly is held at a pressure less than a predetermined pressure to guard against over-dilating or over-sizing the balloon within the airway, which can cause unwanted damage to the airway. In the event that the inflow lumen, throttle, or outlet lumen gets clogged, a pressure sensor, such as one positioned on the handle, will detect the buildup of pressure in the corresponding line and will automatically trigger the coolant pump to shut off so as not to rupture the expandable cooling member or over-dilate the balloon, and/or the energy source for the electrode to shut off so that the treatment site does not overheat. In an alternative embodiment, pressure relief valves may be incorporated within the system to alleviate a buildup of pressure at a predetermined or user-controlled pressure limit. Specific pressure measurement assemblies are described in more detail below.

[0074] In one embodiment, and referring to FIG. 13, a pressure sensor **1200** is positioned on or near the handle assembly **1202** and is configured to measure the pressure of the outlet lumen **1204** of catheter shaft **1203**. This embodiment is based on a pressure measurement downstream (i.e. the outlet lumen **1204**) of the ablation assembly **1206**, thereby requiring a theoretical pressure offset to estimate the pressure in the expandable member **1208** of the ablation assembly **1206**. In this embodiment, it is estimated that a pressure in the expandable member **1208** is greater than a pressure at gauge **1200** due to the resistance of flow in the catheter shaft **1203** relative to the expandable member **1208**.

[0075] In yet another embodiment of the invention, a direct measurement of the pressure in the balloon or ablation assembly **1206** is desired so as to eliminate or reduce undetected obstructions or occlusions between pressure sensor **1200** and the expandable member **1208**, which can cause possible over-inflation or rupturing of the expandable member **1208**. Referring to FIG. 14, a pressure sensor **1200** is positioned on or near the handle assembly **1202**. A polymeric and/or metallic tube **1210** is positioned along outflow lumen **1204** of catheter shaft **1203**. A first end **1210a** of tube **1210** is operably coupled to the pressure sensor **1200** while a second end **1210b** terminates within ablation assembly **1206**. Tube **1210** acts as a static column to measure the pressure directly in ablation assembly **1206**; i.e. the pressure through catheter shaft **1203** within tube **1210** is constant such that the pressure measured at pressure sensor **1200** is equal to the pressure in ablation assembly **1206**.

[0076] In this embodiment, tube **1210** can be flexible or rigid. As seen in the figure, a support wire **1214**, such as a Nitinol wire, extends between an interior of a proximal and distal end of expandable member **1208** to provide added axial, torsional, and buckling support for expandable member **1208**. More particularly, a first end **1214a** of support wire **1214** is coupled to the distal end **1203a** of catheter shaft **1203**, while a second end **1214b** is coupled to the distal end of expandable member **1208** at a junction between throttle valve **1209** and expandable member **1208**. In this embodiment, tube **1210** can supplement support wire **1214**.

[0077] In yet another embodiment, and referring to FIG. **15**, a pressure tube **1220** replaces the support wire of the previous embodiments. In this embodiment, tube **1220** is coupled at first end **1220a** to pressure sensor **1200** positioned on or near the handle assembly **1202**. A distal portion **1222** of tube **1220** extends between a proximal and distal end of expandable member **1208**. More specifically, a proximal end **1222a** of distal portion **1222** is coupled to the distal end **1203a** of catheter shaft **1203**, while a distal end **1222b** of distal portion **1222** is coupled to the distal end of expandable member **1208** at a junction between throttle valve **1209** and expandable member **1208**. Distal portion **1222** of tube **1220** includes one or more apertures **1224** positioned perpendicular to a longitudinal axis of tube **1220**. Similar to the previous embodiment, tube **1210** acts as a static column to measure the pressure directly in ablation assembly **1206**; i.e. the pressure through catheter shaft **1203** within tube **1210** is constant such that the pressure measured at pressure sensor **1200** is equal to the pressure in ablation assembly **1206**. Furthermore, tube **1220** provides axial, torsional, and buckling support for the length of catheter shaft **1203** and expandable member **1208**, allowing for thinner and more flexible fluid containment tubes (e.g. inflow and outflow lumens). [0078] Process control loops including the real-time measurements of pressure, flow, and/or temperature within the system coupled with system response to automatically shut off provides a safety mechanism to guard against undesired tissue or airway injury.

[0079] Referring back to FIGS. **6** and **7**, handle assembly **104** is releasably yet fixedly coupled to the working channel of a bronchoscope via a bronchoscope positioning assembly **1012**. In an embodiment, positioning assembly **1012** comprises a scope coupler **1014** shaped so that it is mountable to the working channel in only a single rotational orientation. Scope coupler **1014** is coupled to the bronchoscope (or bronchoscope adapter assembly described below) via spring loaded jaw lever **1016**, a jaw return spring (compression spring) **1018**, sliding jaw **1020**, stationary jaw **1022**, and a silicone seal **1023** optionally bonded to either the stationary jaw **1022** or the coupler **1014**. Force applied to lever **1016** extends spring **1018**, thereby sliding jaw **1020** and jaw **1022** away from locking abutment with a recessed portion of the working channel to release scope coupler **1014** therefrom.

[0080] Spindle tube **1024** is fixedly coupled in both rotational and axial directions to scope coupler **1014** at a first end **1024a**. More particularly, a flange or skirt **1026** of spindle tube **1024** abuts a flat surface **1028** of scope coupler **1014**. A protrusion **1030** of scope coupler **1014** extends into an opening **1032** of spindle tube **1024**, in a mortise and tenon fashion. A tab **1034** clicks into a portion of protrusion **1030** to lock scope coupler **1014** to spindle tube **1024**. A spindle cover **1036** is coupled to handle cover **1002**, such that spindle tube **1024** is axially slidable along its length through spindle cover **1036**.

[0081] Spindle tube **1024** is axially extendable into and out of handle housing **1002** for fine axial adjustment via axial translation of handle housing **1002**, and therefore manifold **1008** and catheter shaft **102**, along longitudinal axis A with respect to spindle tube **1024**. More particularly, spindle tube **1024** has an inner diameter larger than an outer diameter of catheter tube **1004** and therefore shaft **102**, such that catheter tube **1004** and shaft **102** telescope axially within and out of spindle tube **1024** to achieve axial adjustment of ablation assembly **106**. A length of travel of shaft **102** with respect to spindle tube **1024** is chosen based on a desired length of travel of ablation assembly **106** with respect to the working end or tip of the bronchoscope. In an embodiment, as handle housing **1002** extends axially away from scope coupler **1014**, manifold **1008**, being fixed to

housing **1002**, also extends away, pulling shaft **102** with it, thereby retracting the ablation assembly **106** coupled to a distal portion of the shaft **102** toward and optionally entirely or partly into the bronchoscope, and vice versa upon axial translation toward scope coupler **1014**.

[0082] Handle housing **1002** and catheter tube **1004**, and therefore shaft **102**, are prevented from being completely pulled away from spindle tube **1024**. More particularly, a second end **1024b** of spindle tube **1024** is axially retained within housing **1002** via a spindle retainer assembly **1040**. Spindle retainer assembly **1040** includes a spindle end cap **1042** affixed to an end of spindle tube **1024**, and a handle stop **1044** coupled to spindle end cap **1042** via a U-shaped pin. As handle housing **1002** is axially extended from scope coupler **1014**, in its full extension, radial protrusions of handle stop **1044** abut radially extending features protruding from an interior surface of handle housing **1002**, thereby prohibiting further axial extension. Similarly, features on an interior surface of handle housing **1002** along the longitudinal length of housing **1002** about radially extending protrusions on an exterior surface of handle stop **1044** to inhibit handle stop **1044** from rotating within housing **1002**, while spindle end cap **1042** rotates with spindle tube **1024**.

[0083] Handle stop **1044** further includes a notch for releasably engaging manifold **1008** when adjacent to each other in a fully retracted or closed configuration to lock spindle tube **1024** in axial translation with respect to manifold **1008** and housing **1002** if desired.

[0084] A desired length of axial travel of ablation assembly **106** with respect to the working end or tip of the bronchoscope is configured or determined by a total length of shaft **102** of catheter assembly **101**, and the relationship of catheter tube **1004** with respect to spindle tube **1024** of the handle assembly **104**. In one embodiment, the desired length of travel of ablation assembly **106** with respect to the working end or tip of the bronchoscope is equal to or greater than a longitudinal length of ablation assembly **106**. In an embodiment, a longitudinal length of ablation assembly **106** includes a length of an expandable member (e.g. balloon) in addition to the conduit/electrode as seen in FIG. 7; in an alternative embodiment, the longitudinal length of ablation assembly **106** includes the length of the expandable member only. For sake of simplicity, a longitudinal length of ablation assembly **106** is referred to generally, and can incorporate either length.

[0085] To demonstrate the relationship, in a non-limiting embodiment, a working length of a commercially available bronchoscope, measured from the proximal opening of the working channel (to which the handle assembly is couplable) to the distal working end or tip (see FIG. 16) is from about 40 to about 80 cm, more particularly from about 50 to about 65 cm, and more particularly from about 55 to about 60 cm. A portion of shaft **102** extending within housing **1002** of handle assembly **101** can be about 15 to about 20 cm in length. A total length of shaft **102** of catheter assembly **101**, including the portion within housing **1002** and the length extending through the bronchoscope (but not including the length of ablation assembly **106**), is about 55 to about 100 cm, more particularly about 65 to about 85 cm, and more particularly about 70 to about 80 cm. This ensures that shaft **102** can extend through and out of handle housing **1002**, and through and out of the bronchoscope when handle assembly **104** is coupled to the bronchoscope. In a particular embodiment, shaft **102** includes additional length such that a distal portion of shaft **102** having ablation assembly **106** coupled thereto is extendable beyond the working end or tip of the bronchoscope such that about 0.1 cm to about 4 cm of shaft **102** is exposed distally of the bronchoscope.

[0086] Further, in this embodiment, an ablation assembly **106** comprises a balloon or basket having a longitudinal length from about 1 to about 8 cm, more particularly from about 3 to about 5 cm, and more particularly from about 4 cm. With spindle tube **1024** fixed to scope coupler **1014**, handle housing **1002** can travel axially a distance selected so that catheter tube **1004**, and therefore a portion of shaft **102**, telescopes within or axially translates with respect to spindle tube **1024** of the handle assembly **104** a length of equal to or greater than a longitudinal length of ablation assembly **106**. This allows ablation assembly **106** to shift between being fully extended out of the working end or tip (with or without a gap outside of the working end depending on the total length of shaft

102) of the bronchoscope to fully retracted within the working end of the bronchoscope (with or without a gap inside of the working end depending on the total length of shaft **102**).

[0087] For example, a total length of shaft **102** may be selected such that, when handle assembly **104** is coupled to the bronchoscope and is fully closed or collapsed (e.g. catheter tube **1004** is at maximum nesting position within spindle **1024**, herein referred to as “fully closed handle configuration”), a proximal shoulder of the balloon of ablation assembly **106** extends from about 0.1 to about 4.0 cm, and more particularly about 2 cm, beyond the working end of the bronchoscope. When handle assembly **104** is in a “fully extended handle configuration” in which housing **1002** is at a maximum axial distance from scope coupling assembly **103**, ablation assembly **106** entirely retracts within the working end to allow for unobstructed visualization of the airway. In this embodiment, the length of travel is greater than the length of the balloon, such as, for example, the length of travel is from about 1 cm+0.1-4.0 cm to about 8.0 cm+0.1-4.0 cm. Optical coupling between the viewing device of the bronchoscope and the balloon occurs at a partial axial extension of handle assembly **106** (i.e. “partially extended handle configuration”) in which the proximal shoulder of the balloon abuts the working end of the bronchoscope.

[0088] In an alternative embodiment, when handle assembly **104** is in the fully closed handle configuration, a proximal shoulder of the balloon of ablation assembly **106** abuts the working end of the bronchoscope, and is optically coupled to a viewing device positioned on the working end of the bronchoscope. Catheter tube **1004**, and therefore a shaft **102**, axially translates with respect to spindle tube **1024** at least a distance equal to length of the balloon such that upon full axial extension of handle assembly **104** to the fully extended handle configuration, the entire balloon is retracted into the working end of the bronchoscope.

[0089] In another alternative embodiment, when handle assembly **104** is in the fully extended handle configuration, the proximal shoulder of the balloon of ablation assembly **106** abuts the working end of the bronchoscope, and is optically coupled to a viewing device positioned on the working end of the bronchoscope. In this embodiment, ablation assembly **106** is not retractable within the bronchoscope.

[0090] In a particularly aspect, the length of travel of handle housing **1002**, and hence ablation assembly **106**, is from about 1 to about 20% of the total length of shaft **102**, and more particularly from about 1 to about 10% of total length of shaft **102**. In preferred embodiments, the length of travel of handle housing **1002** relative to spindle tube **1024**, and hence ablation assembly **106**, is at least the length of ablation assembly **106**, preferably being at least about 105% of the length of ablation assembly **106**, and more preferably at least about 110% of the length of ablation assembly **106**.

[0091] In a particularly aspect, spindle tube **1026**, catheter tube **1004**, or other components of handle assembly **104** can comprise indicator notches or tabs markings, sensors, lighted indicators, or other suitable devices at predetermined position(s) to provide a visual and/or tactile response when the balloon of ablation assembly **106** is physically abutting the working end of the bronchoscope to indicate optical coupling of the viewing device and the ablation assembly, when the ablation assembly has been fully retracted into the bronchoscope, and/or at any of a variety of other desired locations.

[0092] Handle housing **1002**, and therefore manifold **1008**, catheter tube **1004**, and catheter shaft **102**, are also rotatable with respect to spindle tube **1024** by rotation of handle housing **1002** with respect to the bronchoscope and scope coupler **1014** for fine circumferential adjustment of the ablation assembly within a treatment area. One or more bearings **1038a, b** are frictionally affixed to an interior surface of handle housing **1004** to aid smooth rotation of handle housing **1002** about spindle tube **1024**. Optionally, because handle stop **1044** rotates with housing **1002** about spindle tube **1024** and spindle end cap **1042**, handle stop **1044** and/or spindle tube **1024** can include indentations that provide a tactile indication or “clicks” corresponding to an amount of rotation. For example, a click can correspond to 1 or more degrees of rotation from a neutral position.

[0093] As described in the Background section, a fiber optic camera portion of the bronchoscope is fixed in relation to the bronchoscope, such that as the bronchoscope and catheter assembly rotates, the image portrayed remains right side up. This can cause disorientation as to the actual orientation of the ablation assembly within a treatment site, such as an airway. Referring now to FIGS. **16** and **17A-C**, in an embodiment, a distal end **2000** of shaft **102** can comprise one or more markers **2002** for indicating an orientation of ablation assembly **106** within a treatment site. In an embodiment, marker **2002** comprises a longitudinal stripe or band, such as a black pad printed band, that longitudinally extends along at least a portion of an exterior of distal end **2000**. When the scope coupling assembly **103** and handle assembly **104** are coupled to the bronchoscope, elongate shaft **102** of the catheter assembly **101** is fed through the working channel **201** and through the scope **200**, and extends out of the working channel **201** at a working end **200a** of scope **200** as depicted in FIG. **16**. Because the handle assembly can only be coupled to working channel **201** in one, unique orientation, elongate shaft **102** extends through and out of working channel **201** in a known or initial orientation. In this embodiment, elongate shaft **102** is coupled to the handle assembly such that marker **2002** extends through the bronchoscope such that it is aligned with the center of camera **2004**.

[0094] Optionally, radioopaque markers (not shown) can be printed or otherwise deposited on the ablation assembly, such as near or on an electrode, such that the orientation of the ablation assembly can be viewed by radiography, fluoroscopy, ultrasound, or other quick confirmation scan.

[0095] Band **2002** can also aid in the axial orientation of the ablation assembly with respect to the working end of the bronchoscope. For example, once the ablation assembly is expanded in the airway, it can be retracted closer to the bronchoscope until the band is no longer visible by the camera. This indicates an optimal distance between the balloon or expandable member and the camera such that optical coupling of the ablation assembly is accomplished. This allows the electrode of the ablation assembly to be viewed from the working end of the bronchoscope. The camera can then be moved independently of the ablation assembly.

[0096] Braiding **2006** can be incorporated along at least a portion of distal end **2000** of shaft **102**. Braiding **2006** gives torsional stability to distal end **2000** so that steering of housing **1002** and manifold **1008** of handle assembly **104** is translated the entire length of shaft **102** so that ablation assembly **106** rotates and axially translates directly in response to handle and/or bronchoscope movements. Braiding **2006** also prevents or inhibits kinking of shaft **102**. A small gap of unbraided section **2008** can be included on the end of distal end **2000** so as not to interfere with optical coupling, as described above.

[0097] Referring back to FIGS. **5A-5C**, once engaged with device **200**, a user can maneuver catheter assembly **101** with ablation assembly **106** in the cavity or conduit, such as the airway, in which treatment is taking place. Coarse adjustment of ablation assembly **106** is accomplished via axial and rotational movement of the device **200** such that handle assembly **104**, and therefore manifold **1008** and catheter tube **1006**, moves with scope coupling assembly **103** and device **200** causing ablation assembly **106** to translate axially and/or rotate.

[0098] Fine adjustment of ablation assembly **106** is accomplished via axial and rotational movement (**220** and **230**) of handle housing **1002** with respect to device **200** and scope coupling assembly **103** as depicted in FIGS. **5A-5C**. For example, moving housing **1002** to a retracted axial position relative to delivery device **200** (FIG. **5B**), such that catheter tube **1004** is nested within spindle tube **1024**, provides maximum extension of the distal end of shaft **102** with ablation assembly **106** relative to the distal portion of device **200** and into the cavity, conduit, or vessel in which the treatment is taking place. This can allow, for example, the user to target a greater range of tissue for treatment. Extension of handle housing **1002** (FIG. **5A**), in which catheter tube **1004** is retracted from spindle tube **1024**, causes the distal end of shaft **102** and therefore ablation assembly **106** to retract back toward the distal portion of device **200**, and in some instances, within the working end of device **200**.

[0099] In some embodiments, and referring to FIG. 5C, handle housing **1002** functions as a control or steering mechanism for adjusting the circumferential or rotational position of shaft **102** and/or ablation assembly **106**. For example, handle housing **1002**, and therefore manifold **1008** and catheter tube **1006**, can be configured to rotate in predetermined degree increments or continuously with respect to the scope coupler **1014** and device **200**. Any of a variety of rotational graduations can be visually and/or tactically (e.g. clicking) incorporated on handle **104**, e.g. 1, 5, 10, 20, 30, 45, 90, 120, or 180 degree increments.

[0100] In an alternative embodiment, depicted in FIG. 18, handle assembly **304** includes a control mechanism for circumferential/rotational maneuverability including a body portion **310** configured to rotate about a spindle tube **305** fixed to scope coupler **303**, such that when the body portion **310** is rotated circumferentially in one direction, the catheter shaft and ablation assembly rotate circumferentially in the same direction, while device **200**, working channel port **202**, and scope coupler **303** are fixed. In other embodiments, body portion **310** is also configured to move axially with respect to spindle tube **305** and scope coupler **303**, such that when body portion **310** is moved axially in one direction, the catheter shaft and ablation assembly move axially in the same direction with respect to device **200** and scope coupler **303**.

[0101] In alternative embodiments, as depicted in FIG. 19, handle assembly **402** includes a first body portion **404** for axial maneuverability and a second body portion **406** for rotational maneuverability. Handle assembly **402** is coupled to working channel port **202** of scope **200** via scope adapter **403**. Adapter **403** is both axially and rotationally fixed with respect to port **202**. A spindle tube **405** is axially fixed to adapter **403**. Handle assembly **402** can comprise a first body portion **404** configured to translate axially along spindle tube, which translates to axial maneuverability of an ablation assembly fixed to a distal end of the catheter shaft. Second body portion **406**, spindle tube **405**, and first body portion **404** are configured to rotate about scope adapter **403**, which translates to rotationally maneuverability of the ablation assembly.

[0102] In still other embodiments (not shown), the controls for axial and circumferential maneuverability can comprise one or more motors, hydraulic or pneumatic cylinders, or other powered manipulators with associated buttons, switches, sensors, etc. configured to maneuver the distal portion of the endoscopic device without the need to manually manipulate a mechanism on the handle. In still other embodiments (not shown), the control mechanisms for precise or fine axial and/or rotational adjustment are automated, such as by servomotors.

[0103] Catheter positioning handle assemblies and systems allow for fine adjustment and maneuverability of the catheter shaft and ablation assembly independent of the bronchoscope or other delivery device and accessories coupled to the delivery device, such as cameras and/or lighting sources. For example, the handle can be used to more accurately position the ablation assembly under visualization with a camera or other optical element associated with the bronchoscope through which the catheter is positioned. Preferably the handle will facilitate maneuvering the catheter such that the ablation assembly is visible with the optical element during treatment. In preferred embodiments, the handle will facilitate optically coupling of an expandable member of the catheter assembly with an optical element of the bronchoscope, and a secondary source for visual cues for positioning can be in the form of radiopaque marks or radiopaque indicators on the ablation assembly to indicate fixing the catheter in position of the energy emitter relative to the bronchoscope to maintain such optical coupling during treatment.

[0104] According to a non-limiting embodiment, in use, during treatment for a pulmonary disease (e.g., a targeted lung denervation procedure) a bronchoscope can be positioned in the airway according to a typical bronchoscopic procedure. The user can chose the right or left main bronchi. A catheter assembly including a balloon with one or more electrodes coupled thereto can be folded, deflated, or wrapped on the distal end of the shaft of the catheter. The shaft and ablation assembly are inserted into a working channel of the bronchoscope and the handle assembly and scope adapter are connected such that the handle assembly is secured to the bronchoscope. The catheter shaft will

then be aligned via the band on the distal end of the catheter shaft such that the user will have knowledge of electrode positioning and orientation relative to the airway by this initial positioning. In some embodiments, markings on shaft can be visible from the bronchoscope, which can aid the user during positioning of electrode. Optionally, radiopaque markers can be incorporated for additional confirmation of position.

[0105] The user can position the catheter shaft and ablation assembly within the airway (e.g., right or left bronchi). Coarse adjustments are made by axial and rotational translation of the bronchoscope itself, as the handle moves with the scope. Fine adjustments are made in the airway for treatment (e.g., lung denervating using radiofrequency therapy) using the axial and circumferential controls of the handle that adjust the shaft and ablation assembly relative to the scope (i.e. without moving the scope) to position electrode in a position advantageous for treatment. Once positioned, such as between cartilaginous rings of the airway, the user can inflate the balloon, which in turn causes the electrode(s) to contact the airway wall.

[0106] The positioning of the electrode and the balloon in the airway can be informed using a camera coupled to the bronchoscope, such as a lens with fiber optics and/or charge-coupled device (CCD) chip, which provides visual cues as to the position of electrode and balloon, relative to the anatomy of the airway and the other devices. Visualization of the electrode and balloon can be performed by moving the camera independently of the electrode and balloon. For example, the user can hold the handle of catheter assembly fixed, and then axially move the camera towards the electrode and the balloon without disturbing their position. With the ability to hold the handle in a stationary position, the electrode and balloon of the catheter assembly maintain a fixed position in the airway, thus allowing a user to freely move the camera axially and circumferentially for various visual views. Once the desired position is achieved, the user can perform treatment by powering the one or more electrodes, while simultaneously cycling coolant through the cooling circuit, as described above.

[0107] In some embodiments, such that when a circumferential lesion is desired, and the electrode size is less than the circumference of the airway, the treatment is repeated. For example, after the initial application of energy, the user deflates the balloon at least partially or fully, such as by stopping or slowing the coolant flow to depressurize balloon and the optional conduit having the electrode thereon. The user can then reposition electrode to a different quadrant rotationally and/or axially displaced from the first quadrant of the tissue by rotating and/or moving the bronchoscope (coarse adjustment) and then the control mechanism on the handle (fine adjustment), which in turn moves the electrode. In some embodiments, the electrode can be adjusted more finely in one or both axial and circumferential directions before ablation using the handle without moving the position of the bronchoscope.

[0108] In one particular embodiment, in which the electrode is a quarter of the circumference of the main bronchus, the initial position of the electrode when inserted through the bronchoscope is ventral. When positioned in the left main bronchus, for example, the handle assembly and bronchoscope are then rotated 90 degrees counterclockwise so that the electrode is now positioned in the left lateral quadrant. Fine axial and/or rotational adjustments can be made to the electrode as necessary, and as described above. Confirmation from the camera that the black band is still centered with the camera is made, as well as other optional confirmations. The ablation assembly is pressurized as described above, and energy is supplied to the electrode to ablate the target tissue. Upon completion of treatment, the balloon is at least partly depressurized, and the bronchoscope and handle assembly are together rotated 90 degrees clockwise such that the electrode is now positioned in the ventral quadrant. The treatment is repeated as described. The bronchoscope and handle assembly are then rotated 90 degrees clockwise to the right lateral, and treatment is repeated. Finally, the bronchoscope and handle assembly are rotated 90 degrees clockwise to the dorsal quadrant and treatment is repeated. The catheter assembly can then be retracted into the bronchoscope, and the bronchoscope can be positioned within the right main bronchus, and the

therapy repeated, as needed. This clockwise rotation of treatment allows a practitioner to easily, and single handedly rotate the bronchoscope and ablation assembly about the circumference of the airway.

[0109] In some embodiments, treatment conditions can require the use of vacuum operation of the bronchoscope, such as to clean an area or the bronchoscope tip. In such embodiments, the entire distal portion of the catheter shaft and ablation assembly can be withdrawn into the working channel of the bronchoscope by pulling the handle so that it is in its fully extended position (See, e.g., FIG. 5A). A vacuum device or aspiration lumen can then be extended through the working channel of the bronchoscope into a position to vacuum debris without compromising the position of the bronchoscope.

[0110] Upon completion of treatment, the user can withdraw the entire shaft and ablation assembly of the catheter into the working channel of the bronchoscope by pulling handle **100** to its fully extended position (See, e.g., FIG. 5A), without detachment of the handle from the bronchoscope. This allows the catheter to be out of the view of the bronchoscope camera field of view to facilitate, for example, an assessment of the effectiveness of the treatment. The user can then orient the bronchoscope to the other bronchi for treatment, and the catheter shaft and ablation assembly can be advanced from the bronchoscope working channel into the other bronchi.

[0111] Referring now to FIGS. **20A-20F**, an embodiment of a bronchoscope adapter assembly is depicted. Adapter assembly **1700** is configured to facilitate coupling of the handle assembly of the catheter assembly as previously described to a working channel port **202** of device **200**, the working channel being differently sized or smaller than a port for accommodating coupler assembly **103**. Adapter assembly **1700** generally includes a shroud **1702** and a coupler **1720**. Shroud (or shell) **1702** comprises a housing portion **1704**, a through bore **1706**, and an optional alignment tab **1708**. Coupler **1720** comprises a flange **1722**, a neck portion **1724**, a collar portion **1726**, a through bore **1727**, an alignment slot **1728**, one or more optional partial relief slots **1729**, and a ledge **1730** on an inner surface of coupler **1720**. Flange **1722** may include a tapered face, as depicted in FIGS. **20A-20F**, so as to allow handle **104** to be more easily connected to adapter assembly **1700**.

[0112] As shown in FIG. **20D**, flange **1722** has an inner diameter **D1**, and an outer diameter **D2**. As shown in FIG. **20C**, neck portion **1724** has an inner diameter **D3** larger than inner diameter **D1** of flange **1722** such that at the junction between neck portion **1724** and flange **1722**, an abutment ledge **1723** is formed. Abutment ledge **1723** can cooperate with a retaining flange **F1** in abutting relationship on an end of port **202**.

[0113] Collar portion **1726** has a variable inner diameter **D4** larger than inner diameter **D3** of such that at the junction between neck portion **1724** and collar portion **1726**, an abutment ledge **1730** is formed. Abutment ledge **1730** can cooperate with a retaining flange **F2** in abutting relationship on an end of port **202**, as shown in FIG. **20C**.

[0114] As shown in FIG. **20B**, collar **1726** may include one or more through and/or partial (**1728**, **1729**) relief slots to facilitate expansion and/or contraction which may occur during engagement and/or disengagement of adapter assembly **1700** with port **202**.

[0115] As depicted in FIGS. **20C** and **20F**, coupler **1720** is installed around the outer perimeter or diameter of port **202**, effectively creating a larger interface for coupling assembly **103** of handle **104**. Slot **1728** of coupler **1720** is configured to communicate with tab **1708** of shroud **1702**. To install adapter assembly **1700** on a port **202** of a device **200**, shroud **1702** is first positioned such that bore **1706** is arranged proximate neck portion **1724** of coupler **1720** and such that collar portion **1726** is not constrained, e.g. **D4** is at a resting diameter, as depicted in FIG. **20E**. Adapter assembly **1700** may then be advanced onto port **202**, with collar portion **1726** free to expand (and therefore **D4** free to expand) to allow ledge **1730** to be passed over first retaining flange **F1** on port **202** and into abutment with second retaining flange **F2**. Shroud **1702** may then be moved to a locked position wherein bore **1706** of shroud **1702** is aligned with collar portion **1726** of coupler

1720, as depicted in FIG. **20F**, thereby squeezing collar portion **1726** to minimize **D4** so that it fits securely around port **202**. Adapter assembly **1700** is then engaged with port **202** and a handle assembly **104** of a catheter assembly **101** may be connected via adapter assembly **103** to flange **1722** of coupler **1720**.

[0116] Optionally, slot **1728** may comprise a tapered profile, being wider toward collar portion **1726** and narrower toward flange portion **1722**. Such a tapered arrangement of slot **1728** facilitates easier engagement and/or disengagement of adapter assembly **1700** with port **202**, in that as shroud **1702** is advanced from collar portion **1726** toward flange **1722** of coupler **1720** to disengage adapter assembly **1700** from port **202**, tab **1708** of shroud **1702** advances in the narrowing slot **1728** causing the collar portion **1726** to spread open such that ridge **1730** clears the retaining flange on port **202**, allowing adapter assembly **1700** to be removed from port **202**.

[0117] The dimensions of adapter assembly **1700** may be selected as desired in order to couple with a variety of delivery devices **200**.

[0118] In an alternative embodiment, the shroud does not include an alignment tab (not shown), allowing an alternative coupler **1800**, depicted in FIG. **20D**, configured with no through slots and instead with multiple partial relief slots **1802** that facilitate expansion and/or contraction which may occur during engagement and/or disengagement of adapter assembly **1700** with port **202**.

[0119] In embodiments, a kit **150** can include a catheter assembly **101** and handle assembly **104** as described above, and a set of instructions **152** for using the contents of **150**, such as depicted in FIG. **21**. Kit **150** may also include coupling assembly **103** and/or adapter assembly **1700**, or adapter assembly **103** and/or adapter assembly **1700** may be provided independently of kit **150**. Kit **150** may be comprised of one or more hermetically sealed and sterilized packages. The contents of kit **150** may be provided already assembled as necessary, or the contents may be provided individually and instructions **152** include the steps of coupling together the contents as described herein. The kit **150** and/or the individual contents of kit **150** may be provided by causing the kit and/or contents to be manufactured and made available to a user.

[0120] Instructions **152** can be any of a variety of tangible or intangible media including, but not limited to a written manual, a CD or CD-ROM, CD, CD-ROM, DVD, BluRay, digitally downloadable or viewable on onto a personal device, such as a computer, tablet, smart device, and/or via verbal instruction by a provider of the kit **150**. In another embodiment, instructions **152** for using the assemblies in accordance with the various embodiments described herein are provided, for example, by a manufacturer or supplier of the assemblies, separately from providing the assemblies, such as by way of information that is accessible using the Internet or by way of seminars, lectures, training sessions or the like.

[0121] Embodiments of the present invention thus allow a user to maneuver the catheter shaft and ablation assembly independently from the bronchoscope, and easily and conveniently into a position that facilitates efficient and effective treatment, ultimately improving patient outcomes and decreasing recovery times. Further, the handle assembly coupled to the bronchoscope becomes a pointer or indicator of the ablation assembly within the treatment site to supplement visual cues provided by the bronchoscope viewing device or camera. Finally, the handle assembly coupled to the bronchoscope allows for easy and efficient visual or optical coupling of the bronchoscope viewing device to the ablation assembly.

[0122] The handle and catheter systems according to embodiments can be used to maneuver ablation assemblies comprising expandable members and one or more energy emitters or electrodes as described above, and/or can be designed or configured for axially, rotational, and/or other manipulation of any of a variety of treatment assemblies including needle and/or needleless injection or drug delivery systems such as, for example, for the injection or delivery of neurotoxins, sclerosing agents, any of a variety of agents for the treatment of pulmonary disorders. For example, the handle and catheter system can be configured to axially (advancing and retracting) and/or rotationally manipulate one or more needles or ports within and around the airway. Various non-

limiting examples of assemblies are described in one or more of the patents and applications listed below, all of which are incorporated by reference in their entireties either herein or above: [0123] U.S. Patent No. 8,088, 127 entitled “Systems, Assemblies, and Method for Treating a Bronchial Tree;” [0124] U.S. Patent Application Publication No. 2011/0152955 entitled “Delivery Devices with Coolable Energy Emitting Assemblies;” [0125] U.S. Patent Application Publication No. 2012/0310233 entitled “Systems, Apparatus, and Methods for Treating Tissue and Controlling Stenosis;” [0126] U.S. Patent Application Publication No. 2011/0118725 entitled Non-invasive and Minimally Invasive Denervation Methods and Systems for Performing the Same;” [0127] U.S. Patent Application Publication No. 2012/0302909 entitled “Methods and Systems for Screening Subjects;” [0128] U.S. Patent Application Publication No. 2011/0301587 entitled “System and Method for Pulmonary Treatment;” [0129] U.S. Pat. No. 8,172,827 entitled “Apparatus for Treating Asthma Using a Neurotoxin;” [0130] U.S. Patent Application Publication No. entitled “Method and Apparatus for Controlling Narrowing of at Least One Airway;” [0131] U.S. Pat. No. 8,483,831 entitled “System and Method for Bronchial Dilation;” [0132] PCT Application Publication No. WO 2013/052501 entitled “Apparatuses and Methods for Injuring Nerve Tissue;” [0133] U.S. Patent Application Publication No. 2013/0310822 entitled “Compact Delivery Pulmonary Treatment System and Method for Improving Pulmonary Function;” [0134] U.S. Provisional Patent Application No. 61/746,460 entitled “Methods for Improving Drug Efficacy;” [0135] U.S. Provisional Patent Application No. 61/779,371 entitled “Fluid Delivery System and Method for Treatment;” [0136] U.S. Provisional Patent Application No. 61/876,925 entitled “Systems, Devices, and Methods for Treating a Pulmonary Disease with Ultrasound Energy;” [0137] U.S. Provisional Patent Application No. 61/847,477 entitled “Methods for Protecting the Esophagus During Pulmonary Treatment Procedures;” and [0138] U.S. Provisional Patent Application Nos. 61/799,742 and 61/870,373, both entitled “Systems, Devices, and Methods for Treating a Pulmonary Disorder with an Agent.”

[0139] It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the subject matter hereof in any way. Rather, the foregoing detailed description will provide those skilled in the art with an enabling disclosure for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the subject matter hereof as set forth in the appended claims and the legal equivalents thereof.

[0140] The embodiments above are intended to be illustrative and not limiting. Additional embodiments are within the claims. Although the present subject matter has been described with reference to particular embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the subject matter hereof.

[0141] Various modifications to the subject matter hereof may be apparent to one of skill in the art upon reading this disclosure. For example, persons of ordinary skill in the relevant art will recognize that the various features described for the different embodiments of the subject matter can be suitably combined, un-combined, and re-combined with other features, alone, or in different combinations, within the spirit of the subject matter hereof. Likewise, the various features described above should all be regarded as example embodiments, rather than limitations to the scope or spirit of the subject matter hereof. Therefore, the above is not contemplated to limit the scope of the present subject matter hereof.

[0142] For purposes of interpreting the claims for the present subject matter, it is expressly intended that the provisions of Section 112, sixth paragraph of 35 U.S.C. are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1.-34. (canceled)

35. A method of pulmonary treatment comprising: introducing a catheter into an airway of a patient through a delivery device; securing a handle assembly coupled to the catheter to the delivery device; moving a control on the handle assembly to maneuver a distal portion of the catheter circumferentially to a first circumferential position within the airway; administering a first treatment at said first circumferential position; moving the control on the handle assembly to maneuver the distal portion of the catheter circumferentially to a second circumferential position within the airway; and administering a second treatment at said second circumferential position; wherein the handle assembly remains secured to the delivery device while maneuvering the distal portion of the catheter to the first and second circumferential positions.

36. The method of claim 35 wherein the delivery device remains in a stationary position relative to the airway as the distal portion of the catheter is maneuvered to the first and second circumferential positions.

37. The method of claim 35 further comprising moving the control while the handle assembly remains secured to the delivery device to retract the distal portion of the catheter into the delivery device after the first treatment and before the second treatment.

38. The method of claim 37 wherein the distal portion of the catheter comprises an expandable member, wherein the control is moved such that an entire length of the expandable member is retracted within the delivery device.

39. The method of claim 35 wherein first and second treatments comprise delivering energy from the distal portion of the catheter to a wall of the airway to ablate a nerve trunk along the wall.

40. The method of claim 35 wherein the delivery device comprises an optical channel, the method further comprising viewing the distal portion of the catheter through the optical channel as it is maneuvered to the first and second circumferential positions.

41. The method of claim 40 wherein the distal portion of the catheter has an expandable member coupled thereto, further comprising moving the control to a predetermined position in which the expandable member is optically coupled with the optical channel when expanded.

42. The method of claim 35 further comprising fluoroscopically visualizing an indicator on the distal portion of the catheter while maneuvering the distal portion to the first and second circumferential positions.

43. The method of claim 35 further comprising manipulating the delivery device with the handle assembly secured thereto to rotationally reposition the distal portion of the catheter relative to the airway after administering the first treatment and before administering the second treatment.

44.-49. (canceled)

50. A method of pulmonary treatment comprising introducing a catheter into an airway of a patient through a delivery device, securing a handle assembly coupled to the catheter to the delivery device, and moving a control on the handle assembly to maneuver a distal portion of the catheter axially and circumferentially within the airway into a position to administer pulmonary treatment to a patient, wherein the distal portion of the catheter is maneuverable independently of a visualization device coupled to the delivery device.

51. A method, comprising: providing a catheter assembly to a user, the catheter including: an elongate shaft; and an ablation assembly coupled to a first end of the elongate shaft, the ablation assembly having an energy emitter thereon; providing a handle assembly to the user, the handle assembly coupleable to a second end of the elongate shaft, the handle assembly including: a handle frame; and a control, movable relative to the handle frame and coupled to the elongate shaft such that moving the control moves the ablation assembly in at least one of an axial and a circumferential direction relative to the handle frame; and providing instructions to the user, the instructions comprising: securing the handle assembly to a port of a delivery device; operating the control on the handle assembly so as to move the ablation assembly axially and circumferentially

into a desired position in an airway of a patient; and administering pulmonary treatment to the patient via the ablation assembly.

52. The method of claim 51, further comprising: providing a coupling assembly to the user, the coupling assembly configured to couple the handle assembly and the port of the delivery device, the coupling assembly configured to remain fixed relative to the delivery device while the ablation assembly is moved.

53. The method of claim 51, wherein providing a catheter assembly to a user comprises causing the catheter assembly to be manufactured and made available to the user, and wherein providing a handle assembly to the user comprises causing the handle assembly to be manufactured and made available to the user.
